Honors: Web Scraping and APIs

Mini Project Report

Eureka

Multi-Source Research Paper and Conference Opportunity Discovery using Web Scraping and REST APIs

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Student Declaration

We hereby declare that the work presented in this document entitled **"Eureka: Multi-Source Research Paper and Conference Opportunity Discovery using Web Scraping and REST APIs"** is an authentic record of design and development work carried out by us under the guidance of Prof. Snigdha Bangal for the Honors course "Web Scraping and APIs". Due acknowledgements have been given in the report for all external components, APIs, and materials used.

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Date: October 12, 2026

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Certificate

This is to certify that the Mini Project Report titled **"Eureka: Multi-Source Research Paper and Conference Opportunity Discovery using Web Scraping and REST APIs"** submitted by **Bhumit Maru** (16014225801) and **Akash Majumdar** (16014225814) from TY AI&DS (Batch B3) is a record of bonafide engineering work carried out under my supervision in partial fulfillment of the requirements for the Honors course "Web Scraping and APIs".

Prof. Snigdha Bangal	**Head of Department**
Course Instructor	Dept. of AI & Data Science

Abstract

Academic research relies heavily on the ability to discover relevant literature and publish in reputable conferences. However, data regarding research papers and Call for Papers (CFPs) is highly fragmented across various publisher databases, pre-print servers, and individual conference websites. The "Eureka" project bridges this gap by providing a unified discovery platform.

This project utilizes a hybrid data-gathering approach, integrating structured data from RESTful APIs (such as arXiv and Semantic Scholar) with unstructured data extracted via Web Scraping from websites that lack public APIs. The aggregated data is sanitized, organized, and presented through an intuitive user interface, allowing students and researchers to execute multi-source queries, filter by publication dates or citations, and track upcoming conference deadlines from a single dashboard. This report details the motivation, lab concepts applied, feature set, system architecture, and user interface of the Eureka platform.

Acknowledgement

We express our sincere gratitude to our course instructor, **Prof. Snigdha Bangal**, for her continuous guidance, constructive feedback, and practical lab sessions throughout the "Web Scraping and APIs" Honors course. Her insights into ethical scraping practices and API integration were instrumental in the development of this project. We also thank the Department of Artificial Intelligence & Data Science for providing the infrastructure and resources necessary to successfully execute and document this mini-project.

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1 INTRODUCTION, MOTIVATION, AND SCOPE

1.1 Introduction

The process of literature review and publication planning is a critical phase for any researcher, student, or academic professional. Currently, users must individually visit platforms like IEEE Xplore, ACM Digital Library, Google Scholar, and arXiv to find relevant papers. Furthermore, tracking upcoming conferences and Call for Papers (CFPs) requires monitoring entirely different academic forums. **Eureka** is a software solution designed to eliminate this fragmentation by serving as a unified search and discovery aggregator built upon advanced web scraping techniques and REST APIs.

1.2 Motivation

The primary motivation behind Eureka is to solve the inefficiency of manual academic searching. Researchers often suffer from "platform fatigue"—spending more time navigating different search syntaxes, export formats, and paywalls than actually reading the papers. Furthermore, many high-quality, domain-specific conferences do not publicize their deadlines via centralized APIs, making them easy to miss. By automating the data retrieval process across multiple domains into one accessible interface, Eureka empowers users to focus on research rather than data hunting.

1.3 Scope of the Project

The scope of the Eureka project includes:

- **API Integration:** Querying open academic databases that provide structured REST APIs (e.g., CrossRef, Semantic Scholar, arXiv).
- **Web Scraping:** Extracting data from academic portals and conference listing sites that do not provide native APIs, utilizing HTML DOM parsing techniques.
- **Data Normalization:** Standardizing diverse JSON responses and scraped HTML text into a uniform data model (Title, Author, Abstract, Year, Link, Deadline).
- **User Interface:** Providing a responsive, search-driven UI where users can input keywords and immediately view aggregated results from all integrated sources.

Out of Scope: Hosting PDF files directly (only links/DOIs are provided) and bypassing authentication mechanisms or paywalls (only open-access data or metadata is scraped/queried).

2 CONCEPTS USED (COVERED IN LAB)

The development of Eureka heavily relied on the practical concepts covered during the Honors Web Scraping and APIs lab sessions. The following core techniques were implemented:

2.1 1. REST API Integration & JSON Parsing

Many modern academic repositories provide RESTful APIs. As taught in the labs, the project utilizes HTTP methods (`GET`, `POST`) to communicate with these endpoints.

- **Requests Library:** Used to construct HTTP requests with appropriate headers (including User-Agent to comply with API policies).
- **JSON Manipulation:** Extracting specific fields from deeply nested JSON responses returned by APIs, handling missing keys, and transforming the data into internal structures.
- **Query Parameters:** Dynamically encoding user search terms and pagination offsets into URL parameters.

2.2 2. Static Web Scraping (HTML Parsing)

For conference websites and repositories lacking APIs, web scraping was employed.

- **BeautifulSoup (bs4):** Utilized for parsing HTML and XML documents. We applied lab concepts such as finding elements by ID, navigating the DOM tree, and extracting text from specific `<div>` or `<span>` classes containing paper titles and conference dates.
- **CSS Selectors:** Used to target precise elements within complex page layouts, ensuring robust data extraction even when site structures contain nested tables.

2.3 3. Ethical Scraping & Rate Limiting

As emphasized in the course, ethical considerations were paramount:

- **Timeouts & Delays:** Implementing ‘sleep()‘ functions between iterative scraping requests to avoid overloading target servers.
- **Robots.txt Compliance:** Checking target domains to ensure the specific paths being scraped are not disallowed by the site administrator.
- **Error Handling:** Using ‘try-except‘ blocks to manage HTTP 403 (Forbidden), 404 (Not Found), and 429 (Too Many Requests) errors gracefully without crashing the application.

3 DESCRIPTION OF EVERY FEATURE

Eureka is designed with a modular feature set that addresses the specific needs of academic researchers.

3.1 Feature 1: Multi-Source Paper Discovery

This is the core feature of the platform. When a user inputs a research topic (e.g., "Federated Learning"), the system concurrently dispatches queries to multiple sources. It fetches structured metadata from REST APIs while simultaneously running scraper scripts on targeted websites. The results are then merged, deduplicated, and presented in a unified list, displaying the Paper Title, Authors, Abstract snippet, Publication Year, and Source Link.

3.2 Feature 2: Conference and CFP Tracking

Instead of just finding past literature, Eureka helps users plan future publications. The platform scrapes prominent Call for Paper (CFP) listing websites to extract upcoming conferences. The data is parsed to identify the Conference Name, Location, Focus Domain, and most importantly, the Submission Deadline.

3.3 Feature 3: Unified Search and Filtering

Because data is aggregated from sources with wildly different formats, Eureka normalizes the data to allow global filtering. Users can sort the combined results by relevance or recency. The search bar supports multi-word queries, and the backend dynamically translates these into the specific syntax required by each individual API (e.g., URL encoding spaces to `%20` or `+`).

3.4 Feature 4: Seamless UI/UX and Result Export

The frontend interface abstracts away all the complexity of APIs and web scraping. Users interact with a clean, responsive dashboard. Additionally, each result contains a direct hyperlink resolving to the paper's DOI or the conference's official homepage, allowing users to exit the platform directly to the source material.

4 USER INTERFACE & SCREENSHOTS

Note: The following sections display the user interface of the Eureka platform, demonstrating the features running as described in the previous section.

4.1 UI: Landing Page and Unified Search

The landing page provides a clean, distraction-free environment for the user to initiate their research query.

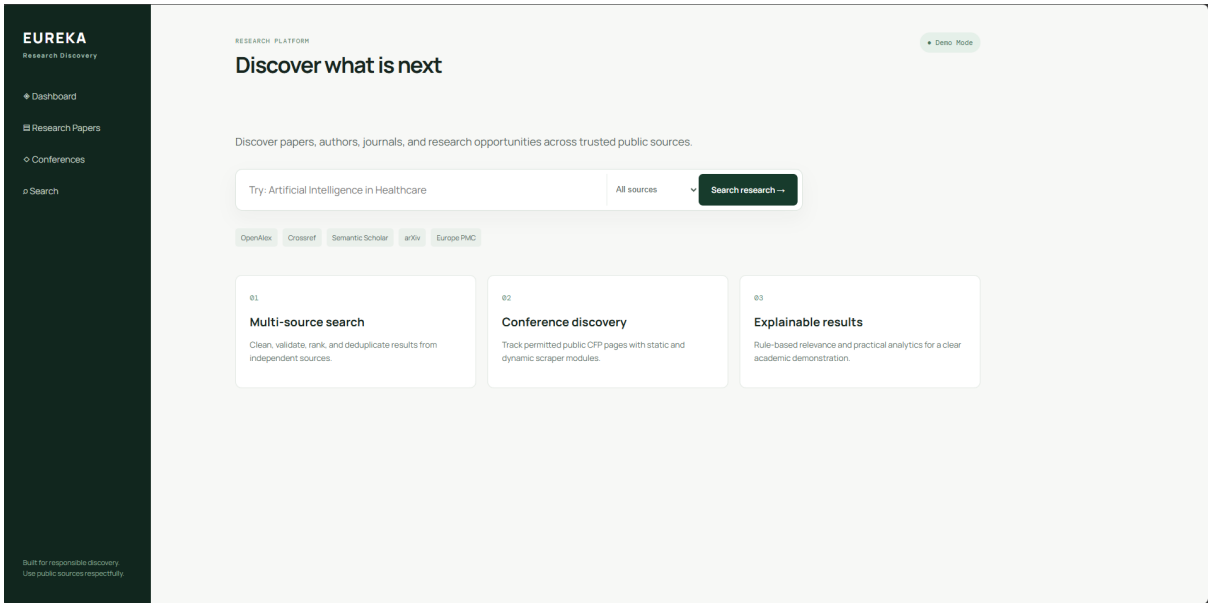


Figure 1: Landing Page demonstrating the Unified Search interface

The UI consists of a prominent search bar where users can input keywords. Radio buttons or dropdowns allow the user to toggle between searching for "Research Papers" via APIs or "Upcoming Conferences" via Web Scraping.

4.2 UI: Multi-Source Paper Results

Once a query is executed, the backend aggregates data and populates the results page.

This screen shows how parsed JSON data and scraped HTML data are mapped into a consistent card-based layout. Each card displays the paper's title, authors, year, and a button linking to the source.

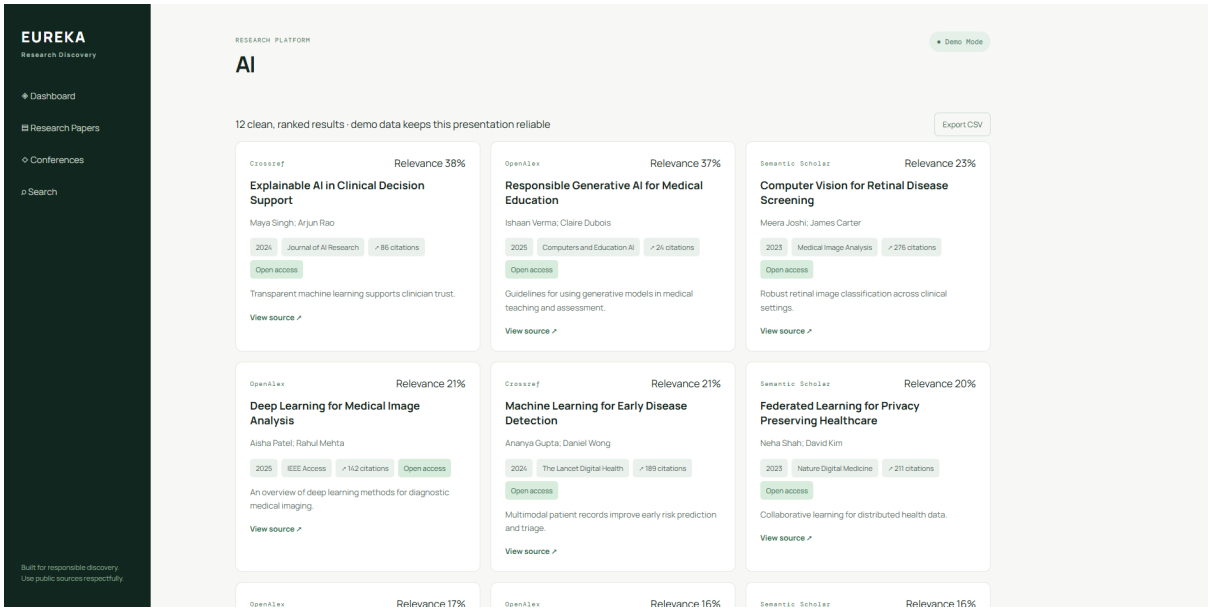


Figure 2: Aggregated Research Paper Results from Multiple Sources

4.3 UI: Conference & CFP Tracker

When the user switches context to find publication opportunities, the application invokes the web scraping modules.

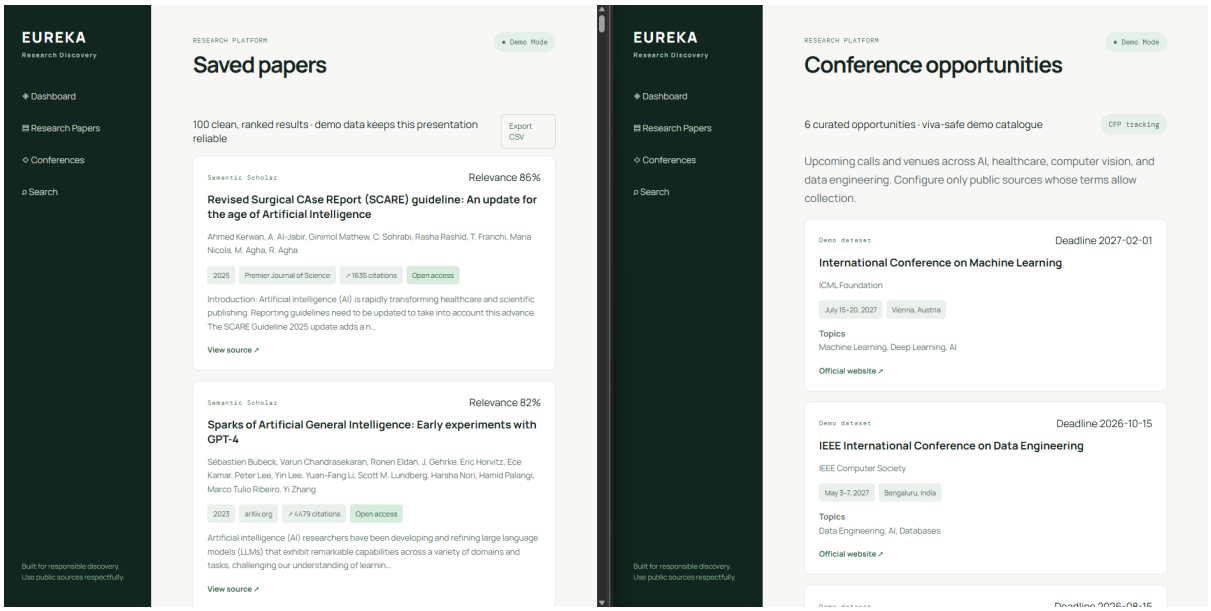


Figure 3: Scraped Conference Data showing Submission Deadlines

This interface displays the scraped conference details. Important metadata such as Submission Deadlines and Event Dates are highlighted, proving the successful extraction of specific DOM elements from third-party CFP websites.

5 SYSTEM ARCHITECTURE & DATA FLOW

5.1 Level 0 Data Flow Diagram (Context Diagram)

The Context Diagram illustrates the high-level interaction between the User, the Eureka system, and the external data sources (APIs and HTML web pages).

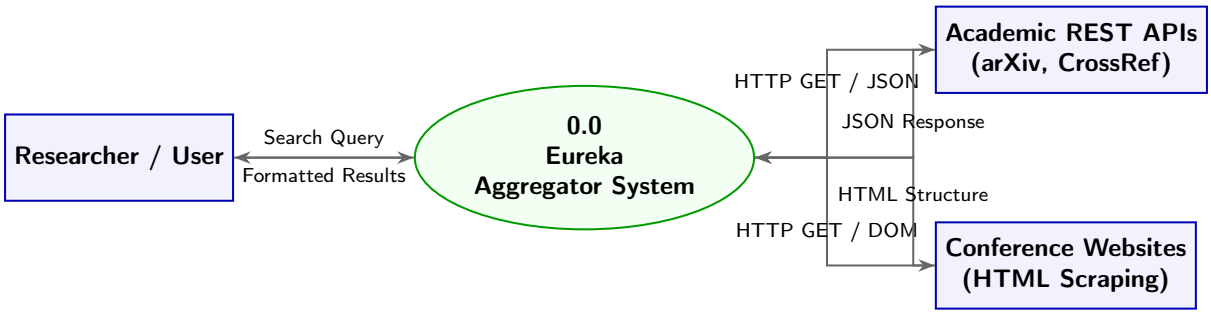


Figure 4: Level 0 Context Diagram

5.2 Level 1 Data Flow Diagram (Detailed Decomposition)

The Level 1 DFD breaks down the internal workings of Eureka, showing how a user query is separated, processed by either the API Handler or the Web Scraper, normalized, and returned to the UI.

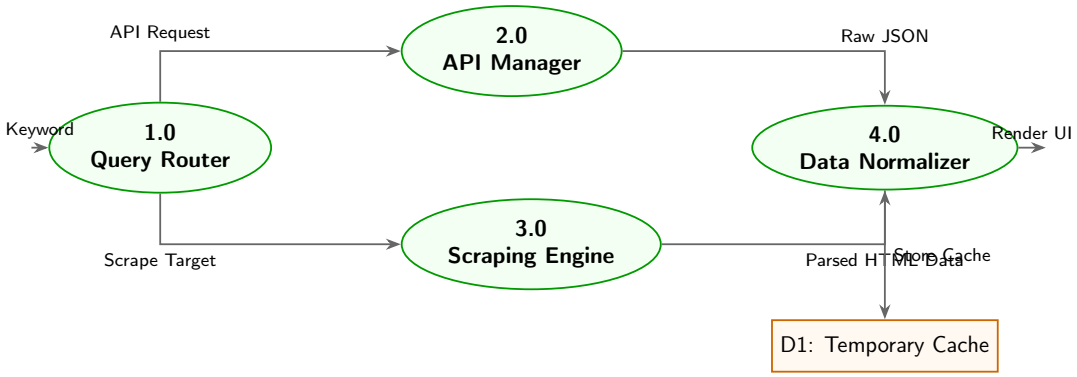


Figure 5: Level 1 Data Flow Diagram

6 CONCLUSION

The **Eureka** mini-project successfully demonstrates the power and utility of combining RESTful APIs and Web Scraping into a single, cohesive application. By applying the concepts learned in the Honors Web Scraping and APIs course—such as HTTP request handling, JSON deserialization, HTML DOM parsing with BeautifulSoup, and ethical scraping practices—we engineered a solution to a real-world problem faced by the academic community.

The system effectively proves that while APIs provide clean, structured, and fast data retrieval, web scraping remains an indispensable tool for accessing information from platforms that lack open developer interfaces (such as many specific conference boards). Eureka successfully bridges this gap, saving researchers significant time by providing a single pane of glass for literature review and publication tracking.

7 FUTURE SCOPE

While the current iteration of Eureka successfully aggregates and displays data, several enhancements can be introduced to expand its capabilities in the future:

1. **Automated Email/Push Notifications:** Implementing a background cron job that periodically scrapes targeted conference websites and alerts users via email when a CFP deadline is approaching.

2. **Citation Exporting:** Adding a feature to dynamically generate and export bibliographic data into standard formats such as BibTeX, RIS, or EndNote directly from the scraped metadata.

3. **Machine Learning Recommendations:** Integrating NLP (Natural Language Processing) to analyze the abstracts of the user's saved papers and automatically recommend similar literature or relevant conferences.

4. **Bypassing Dynamic Content:** Upgrading the static scraper (BeautifulSoup) to a headless browser architecture (like Selenium or Playwright) to successfully scrape modern Single Page Applications (SPAs) that load conference data dynamically via JavaScript.

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